Seana Corners and Dylan Thornburg
ELEN 50

Lab 8 Postlab

Table 2 – Resistance Values

	R_{S1}	R_{f1}	R_{s2}	R_{f2}	
Nominal Values	1k	10k	2k	8k	Ω
Measured Values	0.98526k	10.081k	2.0830k	7.9619k	Ω

2. With the measured values of Rs1, Rf1, Rs2, and Rf2, calculate the expected gains of inverting and non-inverting amplifiers in Figure 2, and record the calculations in Table 3

3. (inverting) Set the oscilloscope to the DC RMS mode for both CH1 and CH2, and observe the difference between the input and output waveforms.

There was a quotient of about 10 between the two amplitudes (the output amplitude was about ten times more).

5. (non inverting) Set the oscilloscope to the DC RMS mode for both CH1 and CH2, and observe the difference between the input and output waveforms.

There was a quotient of about 5 between the two amplitudes (the output amplitude was about five times more).

Table 3 – Gain of Inverting and Non-inverting Amplifiers

	Expected Gain	Measured Gain (1st half cycle)	Measured Gain (2nd half cycle)	% Error (1st half cycle)	% Error (2 nd half cycle)
Inverting Amplifier	-10.2318	-10.207	-10.3208	-.2424	.8698
Non-inverting Amplifier	4.8223	4.776	4.8756	-.9601	1.1053

7. There is a slight offset in the input and output due to likely computer error. The amplitude for each of them vary between 1.07V and 1.09V.

Table 4 – Voltage Amplitudes for Voltage Follower

R_{dec} (Ω)	v_g (V)	Expected v_{out} (Fig. 4(a)) (V)	Measured vout (Fig. 4(a)) (V)	% error	Expected v_{out} (Fig. 4(b)) (V)	Measured v_{out} (Fig. 4(b)) (V)	% error
5,000	1.07	.8917	0.900	.9346	1.07	1.13	-5.607
3,000	1.07	.8025	0.820	2.1807	1.07	1.13	-5.607
1,000	1.05	.525	0.520	-.9524	1.07	1.13	-5.607
800	1.05	.4667	0.470	.7143	1.07	1.17	-9.346
400	1.05	.3	0.297	-1	1.07	1.17	-9.346
200	1.03	.1717	0.179	4.2718	1.07	1.17	-9.346
100	1.03	.0936	0.107	14.2718	1.07	1.19	-11.215
50	1.03	.0490	0.061	24.3689	1.07	1.19	-11.215
10	1.01	.01	8.4mV	-16	1.07	0.440	-58.8785
9	1.03	.0092	8.4mV	-8.5696	1.07	0.388	-63.7383
8	1.03	8.2mV	7.6mV	-7.0291	1.07	0.355	-66.8224
7	1.03	7.2mV	6.8mV	-5.0264	1.07	0.321	-70
6	1.03	6.1mV	5.6mV	-8.8414	1.07	0.244	-77.1963

5	1.03	5.1mV	4.8mV	-6.3301	1.07	0.200	-81.3084
4	1.03	4.1mV	4.0mV	-2.5243	1.07	0.170	-84.1121
3	1.03	3.1mV	3.6mV	16.8544	1.07	0.133	-87.5701
2	1.03	2.1mV	3.2mV	55.6505	1.07	0.100	-90.6542
1	1.03	1mV	3.2mV	210.9903	1.07	0.055	-94.8598

11. Based on your results, explain the advantage of using the voltage follower for any given resistive load.

The advantage of the voltage follower is that it mostly conserves the initial voltage value. Once you get to a low enough Rdec value though (about 32) the output voltage starts to drop significantly. I think it does this due to being out of the operational amplifier's scope.